جامعة التقنية والعلوم التطبيقية
University of Technology and Applied Sciences
نزوى
Nizwa

College of Computing & Information Sciences

Information Technology Department

Final Examination

Spring, AY 2023/2024

CSOP1207 - Principles of Operating Systems

ITSE102 - Introduction to Operating Systems

Date: Monday, June 3 ,2024		Time: 09:00 AM – 11:00 AM	
Max Marks	40	Time Allocation	2 Hours

Student ID		Student Name	
Section		Level	Diploma Year I

Instructions to the Students
1. Answer the questions in the space provided by using blue or black pen only. You will obtain a zero (0) if you use a pencil for all questions. 2. This exam paper consists of five (5) pages including the front page. 3. Use of mobile phones and any electronic gadgets inside the examination hall are not allowed. 4. ***Copying, Cheating, and any kind of malpractices on the examination are strictly prohibited. Action will be taken as per the "Article 68 of UTAS Academic Regulations".*** 5. Students should bring their own pencils, pens, erasers, rulers, calculators, and any other tools required for the exam. 6. The exam has a total of **20** non optional questions.

Section	Allotted Marks	Obtained Marks
A	12	
B	28	
Total Marks	40	

First Marker	Second Marker

Section A – Multiple Choice Questions (12 X 1 = 12 Marks)

1.	Which one of the following is NOT an Operating System?			
	a)	Windows	c)	Linux
	b)	DOS	d)	Oracle
2.	The interval from the time of submission of a process to the time of completion is termed as:			
	a)	Waiting Time	c)	Throughput
	b)	Response time	d)	Trunaround Time
3.	If Resource Allocation Graph (RAG) contains cycles with single instances of resources, it is considered that _____ in the system.			
	a)	there is a Deadlock	c)	there is a no Deadlock
	b)	there may or may not be Deadlock	d)	None of the above
4.	What are the necessary conditions for deadlock to occur?			
	a)	Mutual exclusion, hold and wait, no preemption, circular wait		
	b)	Mutual exclusion, hold and wait, no preemption, resource allocation		
	c)	Mutual exclusion, hold and wait, resource allocation, circular wait		
	d)	Mutual exclusion, hold and wait, no preemption, resource allocation, circular wait		
5.	Which semiconductor technology is commonly used to implement main memory?			
	a)	DRAM	c)	EPROM
	b)	Flash Memory	d)	SRAM
6.	In which scenario would a process be 'rolled out'?			
	a)	When it completes execution	c)	When there is a system error
	b)	When a higher priority process needs to be loaded	d)	When it needs more memory

7.	What term describes unused memory within a partition that is allocated to a process but not used by it?			
	a)	External fragmentation	c)	Memory leak
	b)	Internal fragmentation	d)	Paging
8.	Which term describes the additional time required for a disk to rotate the desired sector to the disk head?			
	a)	Seek time	c)	Rotational latency
	b)	Transfer time	d)	Access time
9.	Which disk scheduling algorithm processes requests in the order they arrive?			
	a)	Shortest seek-time first scheduling	c)	FCFS Scheduling
	b)	SCAN scheduling	d)	C-LOOK scheduling
10.	How is an access-right represented?			
	a)	<object-name, user-name>	c)	<object-name, rights-set>
	b)	<user-group, rights-set>	d)	<domain-name, object-name>
11.	In the access matrix, what do the columns represent?			
	a)	Domains	c)	Users
	b)	Objects	d)	Operations
12.	What is the maximum number of CPU sockets supported by Windows Server 2016 Standard Edition?			
	a)	32	c)	128
	b)	64	d)	Unlimited

Section B – Analysis/Application Questions (28 Marks)

Q13) What are the different types of Multithreading Models? Explain any Two. [2 marks]

Q14) Consider a set of 5 Processes whose Arrival Time(AT), Burst Time(BT), and Priority (lesser the number, higher the priority) are given below:

Process	Arrival Time	Burst Time	Priority
P1	0	6	3
P2	1	4	5
P3	2	5	2
P4	3	3	4
P5	4	7	1

If the CPU Scheduling policy is Priority Scheduling (Preemptive),

a) Draw a Gantt Chart

b) Calculate Average Turn Around Time (TAT)

c) Calculate Average Waiting Time (WT)

Gantth chart: [2 Marks]

Average Turn Around Time : [1 Mark]

Average Waiting Time : [1 Mark]

Q15) Consider the following snapshot to apply the banker's algorithm to answer the following questions:

Processes	Allocation			Max Needs			Availabe		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

a) Calculate the content of the need matrix. [.5 Mark]

b) Check if the system is in a safe state. [4 Marks]

c) Write the Safe Sequence. [.5 Mark]

Q16) Draw the allocation graph for the following situation:

Consider a system with 4 processes (P1, P2, P3, P4) and 3 resources (R1, R2, R3). Each resource has exactly 2 instances. The assignment and request edges are given as follows:

Assignment Edges – [R1 → P2,R1 → P4,R2 → P1,R2 → P3,R3 → P3,R3 → P4]

Request Edges – [P1 → R1,P2 → R3,P4 → R2] [2 Marks]

Does the above allocation graph contain the deadlock? Justify your answer. [1 Mark]

Q17) Given memory partitions of 200K,400K,150K,350K,and 600K (in order), how would each of the First-fit, Best-fit, and Worst-fit algorithms place processes of 100K,275K,225K and 500K (in order)? Please fill in the following table. [2 Marks]

Process	First Fit	Best Fit	Worst Fit
100K			
275K			
225K			
500K			

Which algorithm makes the most efficient use of memory? Justify. [1 Mark]

Best Fit, because it use the memory in effeicent way and not wasting alot of memory size

Q18) Explain the page replacements of Least Recently Used by using the following memory References: : 6, 7, 8, 9, 6, 7, 1, 6, 7, 8, 9, 1, 7, 9, 6. Use memory with 3 pages. [4 Marks}

No of Hits :

No of Miss:

Q19) We assume a disk with 200 tracks and disk request queue has random requests in it. The requested disks in the order are 55,58,39,18,90,160,150,38,184 starting with a track 100. Calculate average seek length using **FIFO**. [4 Marks]

Diagram :

Average Seek Length :

Q20) Create an Access Control Matrix for a scenario involving university faculty and students in managing courses.Consider the following scenario: [3 Marks]

There are three types of users: Faculty Members (Faculty), Teaching Assistants (TAs), and Students.
There are four courses offered by the university: Computer Science, Mathematics, Literature, and Biology.
Users can have different levels of access to each course, including "Read" (R), "Write" (W), and "Execute" (X).
In this matrix:

- Faculty Members have full access (Read, Write, and Execute) to all courses, allowing them to create, modify, and manage course materials, assessments, and grading.
- Teaching Assistants have Read and Write access to all courses, enabling them to assist faculty with course management tasks such as grading assignments, holding office hours, and moderating discussions.
- Students have Read access to all courses, allowing them to view course materials, assignments, and announcements, but they do not have permission to modify course content.